

Receding-Horizon Target Streams for Learned Fixed-Wing Maneuvering

Anonymous Author(s)

Affiliation

Address

email

Abstract: Fixed-wing maneuvering is not waypoint tracking. A fixed-wing aircraft couples attitude, velocity, lift direction, energy, and actuator bandwidth, so a spatial reference can be close in position while still being flown with the wrong attitude or energy state. We study a compositional alternative built around a frozen, PID-free, quaternion-conditioned reinforcement-learning flight skill that maps target heading, pitch, roll, and airspeed commands directly to throttle, elevator, aileron, rudder, and speed-brake commands. Long-horizon maneuvers are expressed as executable target streams for this skill, and their parameters are selected online by Receding-Horizon Target-Stream Optimization (RH-TSO): short closed-loop rollouts of the frozen policy score candidate streams with geometry, energy, safety, and smoothness costs. On four representative maneuvers, the best stream parameters are task dependent: horizontal and mild three-dimensional curves prefer shorter lookahead and lower target speed, while a 90° vertical pull-up prefers longer lookahead and higher target speed. This improves CTE-P90 by 39%, 2%, 24%, and 47% over a default stream on S-curve, figure-eight, helix, and vertical pull-up tasks, respectively. Energy-aware fine-tuning extends the skill toward vertical arcs, but geometry-aware evaluation shows the current boundary clearly: 60° – 150° vertical arcs are B-grade, while the 180° half-loop remains a failure case.

Keywords: fixed-wing aerial robots, reinforcement learning, quaternion control, target streams, receding-horizon optimization

1 Introduction

A fixed-wing aircraft cannot stop at a waypoint, rotate in place, and continue. Its ability to turn, climb, or invert depends on how attitude redirects lift, how pitch trades speed for altitude, and how much authority remains in the aerodynamic surfaces. This makes the usual robotics abstraction of tracking a geometric curve incomplete. A trajectory may have small cross-track error (CTE) while the aircraft is skidding, losing energy, rotating out of the intended maneuver plane, or approaching a stall.

Classical fixed-wing stacks often address this coupling with nested controllers: a planner emits high-level references, a PID or autopilot inner loop tracks attitude and speed, and safety logic limits the envelope. Such stacks are effective in their design regime, but they expose two limitations for learned maneuver composition. First, the learned component is separated from the actuator-level dynamics that ultimately decide whether an aggressive command is executable. Second, Euler-angle targets become awkward near vertical and inverted attitudes, exactly where loop-like maneuvers stress the representation.

We instead train a reusable flight skill that maps state and target attitude-speed commands directly to actuator commands. The policy receives a quaternion attitude error, body-frame target direction,

38 speed error, aircraft state, and previous action; it outputs throttle, elevator, aileron, rudder, and
39 speed brake. The skill is trained once, then frozen. The planning problem is then shifted from “what
40 actuator sequence should be applied?” to “what executable target stream should be sent to the skill?”

41 This target-stream view is the main interface in the paper. A reference maneuver is converted into
42 local heading, pitch, roll, and target-airspeed commands that the frozen policy can attempt to exe-
43 cute. A fixed conversion is not enough. In the latest target-stream sweep, S-curves, figure-eights,
44 and helices prefer shorter lookahead and lower target speed, while a vertical pull-up prefers the
45 opposite. The optimal direction changes with the maneuver class, so a single hand-tuned stream
46 parameterization is not a stable design point.

47 We introduce Receding-Horizon Target-Stream Optimization (RH-TSO), a short-horizon optimizer
48 over executable target-stream parameters. At each replan step, candidate streams are rolled out
49 through the same frozen policy and dynamics used at execution time. The selected stream mini-
50 mizes a cost over geometry, energy, safety, and smoothness. In the current implementation we use
51 deterministic parallel-shooting lattice search because the target-stream space is low dimensional;
52 the formulation also supports sampling-based optimizers when richer stream parameterizations are
53 needed.

54 The paper makes five contributions.

- 55 1. A PID-free, quaternion-conditioned fixed-wing flight skill that directly maps target head-
56 ing, pitch, roll, and airspeed commands to actuator-level control.
- 57 2. An executable target-stream abstraction that converts long-horizon reference maneuvers
58 into local commands for the frozen skill, avoiding the limitations of static waypoint or
59 static attitude targets.
- 60 3. RH-TSO, a receding-horizon optimization method over target-stream parameters, imple-
61 mented here as deterministic parallel-shooting lattice search in the executable skill space.
- 62 4. An energy-aware vertical extension that improves large-attitude vertical-arc behavior
63 through low-speed, energy-retention, safety, vertical-progress, and replay terms without
64 claiming to solve the 180° half-loop.
- 65 5. A geometry-aware evaluation protocol showing why CTE alone is insufficient for loop-like
66 fixed-wing maneuvers and identifying the current capability boundary.

67 All experiments use a high-fidelity fixed-wing simulator with a six-degree-of-freedom F-16 aerody-
68 namic model. The aircraft model is an experimental substrate, not the paper’s contribution.

69 2 Related Work

70 **Learning and control interfaces for agile flight.** Several CoRL systems succeed by placing
71 learning at an inspectable interface rather than replacing the entire autonomy stack. Deep Drone
72 Racing maps onboard images to a waypoint and desired speed that a planner and controller can ex-
73 ecute [1]. Learning-based waypoint navigation similarly couples learned perception to model-based
74 planning and feedback control [2]. These systems motivate our interface choice, but the executable
75 object is different: a fixed-wing maneuver needs full attitude and speed commands, not only a local
76 position goal.

77 **Trajectory tracking under feasibility limits.** DATT frames quadrotor tracking as a problem of
78 actuation limits, disturbances, and sometimes infeasible references [3]. Learned time-allocation
79 methods for quadrotors also reason about trajectories that must pass a feasibility test before they
80 can be flown [4]. Our setting has a related feasibility issue, but the constraint is evaluated through a
81 frozen fixed-wing skill. RH-TSO does not optimize actuator commands or polynomial trajectories;
82 it optimizes the target stream that the learned skill can execute in closed loop.

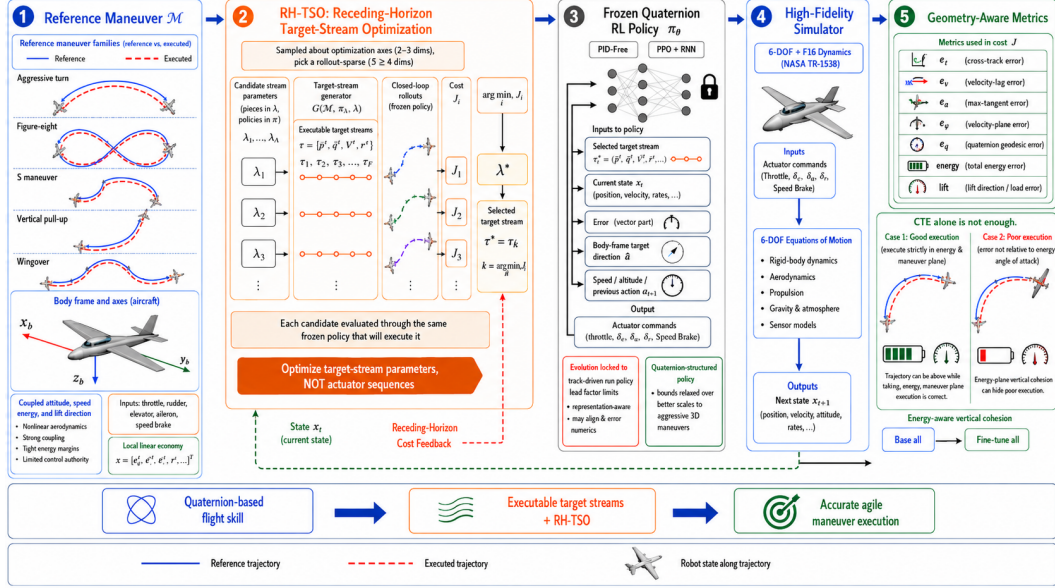


Figure 1: **System overview.** Reference maneuver $\rightarrow$ target-stream generator / RH-TSO $\rightarrow$ local $(\psi, \theta, \phi, v_t^*)$ stream $\rightarrow$ frozen quaternion-conditioned flight skill $\pi_\theta \rightarrow$ actuator commands $(T, \delta_e, \delta_a, \delta_r, \delta_b) \rightarrow$ fixed-wing dynamics $\rightarrow$ geometry-aware evaluation. The optimizer never searches in actuator space. It searches over target-stream parameters and evaluates each candidate by closed-loop rollout through the frozen skill.

Direct learned control. End-to-end swarm controllers show that policies can map local observations directly to thrust commands in multi-robot flight [5]. Learned dynamics and control methods have also been used where classical controller design is difficult, such as soft multicopters [6]. Our skill is direct at the actuator level, but remains conditioned on an explicit target stream. This keeps the interface inspectable while avoiding a hand-tuned inner-loop controller.

Simulation infrastructure and evaluation. Flightmare demonstrates the importance of simulation infrastructure that can trade off speed, fidelity, and learning throughput [7]. Our experiments also rely on high-throughput simulation, but the paper is not a simulator contribution. The evaluation question is specific to fixed-wing maneuvering: position error must be supplemented by attitude, tangent, wing-plane, energy, and safety metrics.

3 Method

3.1 Quaternion-Conditioned Direct-Actuator Flight Skill

The low-level skill is a recurrent actor-critic policy trained with PPO and then frozen for all composition experiments. The policy is PID-free: it directly outputs the five actuator channels used by the simulator. Table 1 summarizes the interface.

Let q be the current aircraft attitude and q^* the target attitude constructed from the target heading, pitch, and roll. The policy observes the shortest-rotation error

$$q_e = q^* \otimes q^{-1}, \quad q_e \leftarrow \text{sign}(q_{e,w})q_e, \quad (1)$$

and uses the vector part of q_e as a bounded three-dimensional attitude error. The corresponding geodesic angle is

$$d_{\text{SO}(3)}(q, q^*) = 2 \arccos(|\langle q, q^* \rangle|). \quad (2)$$

Table 1: Flight-skill interface. The target is local attitude and airspeed, not a waypoint.

Component	Variables
Target command	heading ψ^* , pitch θ^* , roll ϕ^* , target airspeed v_t^*
Attitude observation	vector part of shortest-rotation quaternion error q_e
Direction and speed	body-frame target direction, normalized speed error, normalized speed, normalized altitude
Aircraft state	angular rates, angle-of-attack and sideslip features, previous action
Action output	throttle, elevator, aileron, rudder, speed brake

We adopt quaternion targets to avoid representational discontinuities near vertical and inverted attitudes; the paper does not claim a quantitative Euler-vs-quaternion advantage until the matching ablation table is verified.

The base reward combines attitude tracking, speed tracking, liveness, and altitude/safety terms. The important design choice is not PPO itself, but the reusable skill interface: the policy learns actuator-level tracking of moving attitude-speed targets without a PID or autopilot inner loop.

3.2 Executable Target-Stream Composition

The frozen skill receives a stream of local targets

$$\tau_k = (\psi_k^*, \theta_k^*, \phi_k^*, v_{t_k}^*). \quad (3)$$

For a reference maneuver $\mathcal{M}$ and target-stream parameters λ , a generator G produces

$$\tau_{t:t+H} = G(\mathcal{M}, x_t, \lambda). \quad (4)$$

In the experiments, λ includes lookahead distance, target airspeed, and, for vertical shaping, a pitch offset. These parameters are deliberately higher level than actuators but lower level than a symbolic maneuver label. They determine which future portion of the reference geometry is made executable for the flight skill.

Static waypoint tracking fails for this setting because a point alone does not specify lift direction, attitude preparation, or energy management. Static attitude targets fail for the opposite reason: they do not communicate how the desired attitude should evolve along a maneuver. A target stream provides a moving local contract between the reference geometry and the learned actuator-level skill.

3.3 Receding-Horizon Target-Stream Optimization

RH-TSO solves a short-horizon optimization over target-stream parameters:

$$\lambda_t^* = \arg \min_{\lambda \in \Lambda} J(x_{t:t+H}^\lambda, u_{t:t+H-1}^\lambda, \mathcal{M}), \quad (5)$$

$$\tau_k^\lambda = G(\mathcal{M}, x_k^\lambda, \lambda), \quad (6)$$

$$u_k^\lambda = \pi_\theta(x_k^\lambda, \tau_k^\lambda), \quad (7)$$

$$x_{k+1}^\lambda = f(x_k^\lambda, u_k^\lambda), \quad k = t, \dots, t + H - 1. \quad (8)$$

The optimization variables are target-stream parameters, not actuator commands. This keeps the dimension small and makes each candidate interpretable.

The cost used in our implementation has four terms:

$$J = w_g J_{\text{geo}} + w_e J_{\text{energy}} + w_s J_{\text{safety}} + w_u J_{\text{smooth}}. \quad (9)$$

J_{geo} measures distance to the reference and alignment with its tangent and maneuver plane. J_{energy} penalizes loss of specific energy relative to the maneuver requirement. J_{safety} penalizes low speed,

Algorithm 1 Receding-Horizon Target-Stream Optimization (RH-TSO)

Require: Current state x_t , reference maneuver $\mathcal{M}$, horizon H , frozen policy π_θ

Ensure: First segment of optimized target stream $\tau_{t:t+H}^*$

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1: Sample or enumerate candidate target-stream parameters  $\{\lambda_i\}_{i=1}^N$ 
2: for  $i = 1$  to  $N$  do
3:   Generate candidate target stream:  $\tau_{t:t+H}^{(i)} = G(\mathcal{M}, x_t, \lambda_i)$ 
4:   for  $k = t$  to  $t + H - 1$  do
5:     Compute control:  $u_k = \pi_\theta(x_k, \tau_k^{(i)})$ 
6:     Propagate dynamics:  $x_{k+1} = f(x_k, u_k)$ 
7:   end for
8:   Compute cost:  $J^{(i)} = J(x_{t:t+H}^{(i)}, u_{t:t+H-1}^{(i)}, \mathcal{M})$ 
9: end for
10: Select lowest-cost candidate:  $i^* = \arg \min_i J^{(i)}$ 
11: return First segment of selected target stream:  $\tau_{t:t+H}^{(i^*)}$ 
12: Update  $x_{t+1}$  and repeat
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127 excessive angle of attack or sideslip, over-G events, and altitude violations. J_{smooth} penalizes actu-
128 ator changes.

129 We implement a deterministic parallel-shooting lattice search for its simplicity and grid-optimality
130 in the low-dimensional target-stream space used in our experiments. The same formulation naturally
131 extends to CEM/MPPI-style sampling when the target-stream space becomes higher-dimensional or
132 continuous.

133 3.4 Energy-Aware Vertical Extension

134 The base skill is reliable on horizontal and moderate-attitude targets, but vertical arcs expose energy
135 loss and loop-plane drift. We fine-tune from the base quaternion skill with a curriculum that adds
136 vertical-arc targets while replaying original tasks to retain horizontal behavior. The reward adds
137 low-speed penalties, energy-retention terms, altitude and vertical-progress shaping, alpha/beta/G
138 safety penalties, and action smoothness. Loop-like phases additionally include geometry terms for
139 velocity-tangent, nose-tangent, nose-velocity, and wing-plane alignment.

140 This extension should be read as a capability expansion, not a solved aerobatics result. It improves
141 behavior on 60° – 150° vertical arcs, but the 180° half-loop remains outside the current reliable en-
142 velope.

143 3.5 Geometry-Aware Maneuver Metrics

144 CTE is necessary but not sufficient. For loop-like maneuvers we also evaluate:

$$e_v = \arccos(\hat{v} \cdot \hat{t}_{\text{ref}}), \quad e_n = \arccos(\hat{n}_{\text{body}} \cdot \hat{t}_{\text{ref}}), \quad (10)$$

$$e_{nv} = \arccos(\hat{n}_{\text{body}} \cdot \hat{v}), \quad e_w = \arccos(|\hat{w}_{\text{body}} \cdot \hat{n}_{\text{plane}}|), \quad (11)$$

$$e_q = d_{\text{SO}(3)}(q, q_{\text{ref}}). \quad (12)$$

145 Here e_v is velocity-tangent error, e_n is nose-tangent error, e_{nv} is nose-velocity error, e_w is wing-
146 plane error, and e_q is quaternion geodesic error. These metrics are reported together with airspeed,
147 alpha/beta, G-load, altitude, and energy envelopes.

148 4 Experimental Setup

149 The simulator uses a six-degree-of-freedom fixed-wing dynamics model with aerodynamic lookup
150 tables, actuator limits, and 50 Hz physics integration. The RL policy is queried at a lower control
151 rate and outputs normalized actuator commands that are decoded to the simulator channels.

Table 2: Target-stream selection from the latest closed-loop sweep. Best values are compared with the default stream (1000 m, 250 m/s). CTE-P90 is in meters.

Maneuver	Best (L, v_t^*)	Best CTE-P90	Default CTE-P90	Improvement
S-curve	(600, 220)	918	1503	39%
Figure-eight	(600, 220)	1017	1039	2%
Helix / mild-3D	(600, 220)	524	691	24%
90° vertical pull-up	(1500, 280)	51	96	47%

The target-stream experiments use the frozen energy-aware skill. We evaluate S-curve, figure-eight, helix/mild-3D, and 90° vertical pull-up tasks. The default target-stream setting is $(L, v_t^*) = (1000 \text{ m}, 250 \text{ m/s})$. The reported lattice compares this default with task-adaptive selections from the same closed-loop rollout evaluation.

The loop-quality evaluation covers vertical arcs from 60° through 180°. The manuscript currently treats the detailed loop-quality numeric table as a verification item; the only claim made from the present diagnosis is the qualitative frontier: 60°–150° arcs are B-grade, and 180° half-loop is Fail.

5 Results

5.1 Low-Level Skill Tracking

The direct-actuator skill can track moving attitude-speed targets without a PID inner loop, which makes it reusable as the executor for target streams. The final manuscript should include a compact tracking table with attitude geodesic error, speed error, survival, and safety violations for held-out target schedules. Those numbers are not restated here because the supporting evaluation log is not present in the manuscript directory.

5.2 Euler vs. Quaternion Target Encoding

Quaternion conditioning is used for a representational reason: the error in Eq. (1) remains well defined near vertical attitudes and through inverted phases. This is sufficient motivation for the current manuscript. A quantitative claim that quaternion encoding outperforms Euler encoding should be added only after the Euler baseline table is verified.

5.3 Target-Stream Ablation

Static spatial targets and static attitude-speed targets are not executable descriptions of long fixed-wing maneuvers. The target stream supplies a time-varying local command that communicates both the desired attitude and how that attitude should evolve along the reference. This ablation should be visualized with representative trajectories once the missing figures are added.

5.4 Receding-Horizon Target-Stream Selection

The strongest current evidence for RH-TSO is the task dependence of the selected stream parameters. Table 2 reports the latest sweep over a two-dimensional target-stream slice. Horizontal and mild three-dimensional maneuvers prefer shorter lookahead and lower target speed; the vertical pull-up prefers longer lookahead and higher target speed. Therefore the target stream cannot be fixed once for all maneuvers.

The result is not that the particular pair (600, 220) is universally good. It is good for horizontal and mild 3D curves and wrong for the vertical pull-up. Conversely, the vertical choice (1500, 280) is a poor fit for the curved tasks. This is the empirical reason to optimize executable target-stream parameters online.

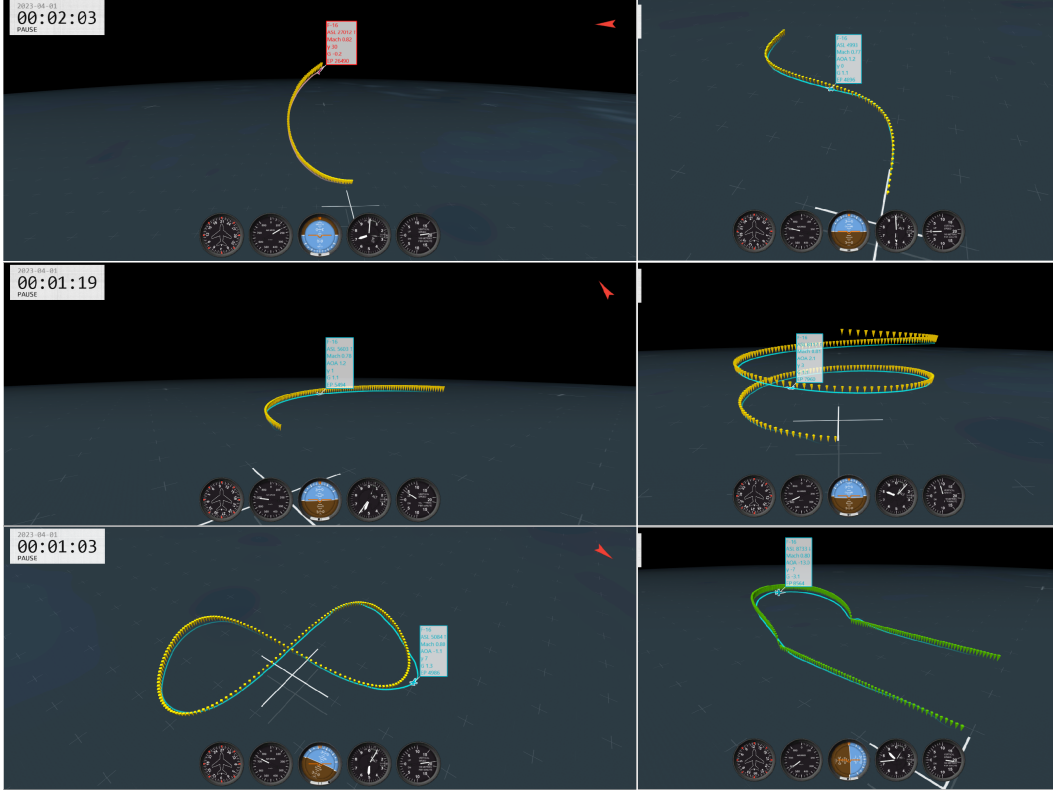


Figure 2: Representative maneuvers.

Table 3: Loop-quality frontier. Metric values are intentionally left as verification placeholders; grades reflect the current strict diagnosis.

Arc angle	CTE	Vel.-tan.	Nose-tan.	Wing-plane	Grade
60°	verify	verify	verify	verify	B
90°	verify	verify	verify	verify	B
120°	verify	verify	verify	verify	B
150°	verify	verify	verify	verify	B
180°	verify	verify	verify	verify	Fail

186 5.5 Energy-Aware Vertical Extension

187 Energy-aware fine-tuning extends the base skill toward larger-attitude vertical arcs by explicitly
 188 shaping low-speed behavior, energy retention, vertical progress, safety margins, and original-task
 189 replay. The current evidence should be reported as a frontier rather than a solved-loop result. The
 190 skill reaches B-grade behavior on 60°–150° vertical arcs under loop-quality metrics, while the 180°
 191 half-loop remains Fail.

192 5.6 Geometry-Aware Evaluation and Capability Boundary

193 Table 3 records the current capability boundary without inventing unavailable metric values. The
 194 detailed numeric cells must be filled from loop-quality logs before submission.

195 The important evaluation lesson is qualitative but consequential: CTE alone can rank a loop-like ma-
 196 neuver incorrectly. A trajectory close to the reference path can still have the wrong wing plane, nose

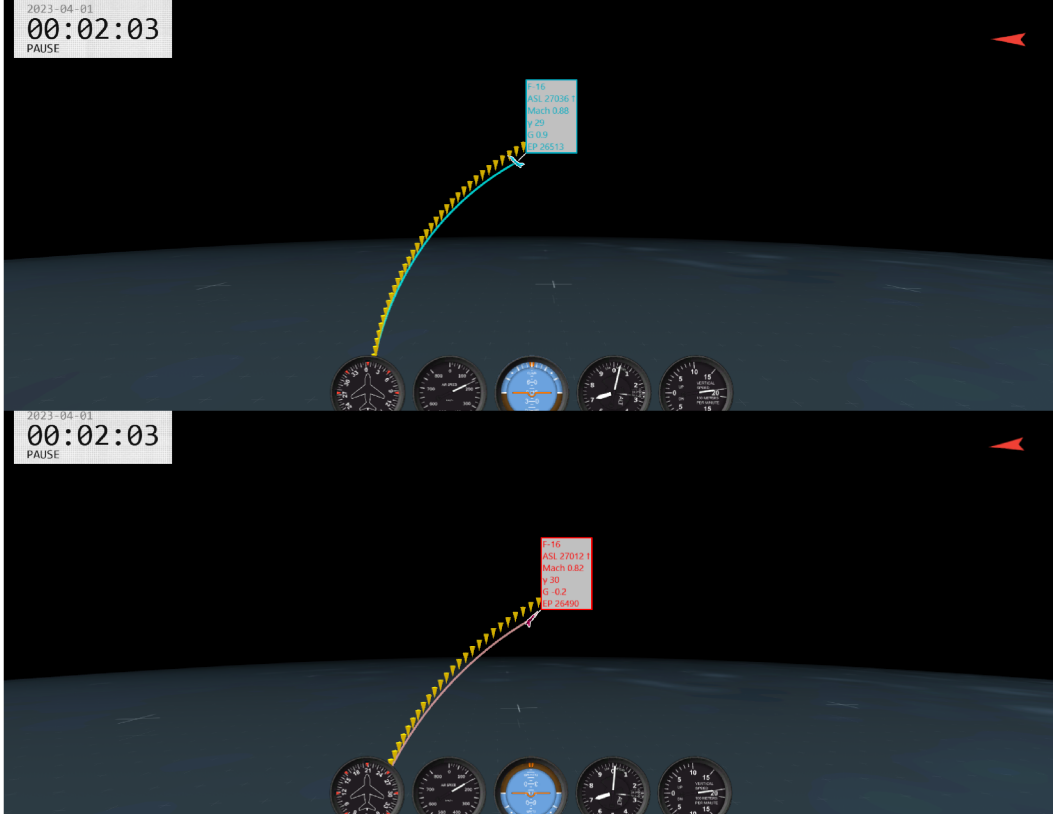


Figure 3: CTE-only tracking can hide physically invalid loop geometry. **Top:** a trajectory with small positional error is nevertheless invalid because the aircraft develops an excessive angle of attack and effectively stalls. **Bottom:** the proposed geometry-aware criterion favors trajectories that are not only close to the reference path, but also align the nose and velocity directions with the path tangent, producing a physically plausible maneuver.

197 direction, velocity tangent, or energy state. The geometry-aware metrics reveal the true boundary of
 198 the current single-skill system.

199 6 Discussion and Limitations

200 **Why target-stream optimization?** The target-stream space is compact, interpretable, and exe-
 201 cutable. It is compact because a few parameters can change the local stream over a horizon. It is
 202 interpretable because selected lookahead and speed reveal how the skill prepares for a maneuver. It
 203 is executable because every candidate is evaluated through the frozen policy and dynamics before
 204 selection.

205 **What RH-TSO does not claim.** The deterministic lattice is grid-optimal only over the chosen
 206 candidate set. It is not globally optimal, not actuator-space MPC, and not evidence that lattice
 207 search is better than CEM or MPPI in a richer continuous space.

208 **Simulation scope.** All results are in simulation. Real deployment would require state-estimation
 209 noise, wind, actuator dynamics, airframe mismatch, and safety monitors that are outside this paper.

210 **Capability boundary.** The present system is not a full aerobatics solution. Horizontal and mild
 211 3D maneuvers are reliable under the tested target streams; 60° – 150° vertical arcs are B-grade under

loop-quality evaluation; the 180° half-loop is a failure boundary. A preliminary phase-gated residual specialist may be a promising direction for the inverted-transition region, but it is outside the main evaluation because this paper focuses on reusable target-stream composition with a single frozen skill.

7 Conclusion

We presented a fixed-wing maneuvering framework organized around a reusable, PID-free, quaternion-conditioned flight skill and executable target streams. RH-TSO selects target-stream parameters by short closed-loop rollouts of the frozen skill, which exposes why fixed stream parameters fail across maneuver classes: curved horizontal tasks prefer short lookahead and lower speed, while vertical pull-up prefers longer lookahead and higher speed. The same evaluation also clarifies the current boundary. Energy-aware fine-tuning improves vertical arcs, but geometry-aware metrics show that 180° half-loop remains Fail rather than solved. The next step is to replace the remaining verification placeholders with final loop-quality tables and representative trajectory figures.

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